Report

Analysis of Quantization Results

Overview of Metrics

Quantization techniques aim to reduce the model size while maintaining acceptable levels of performance and latency. Key metrics for evaluating quantization techniques include:

1. **Perplexity**: Measures the model's prediction capability; lower values are better.
2. **Inference Latency**: Reflects the time required for processing; lower is better.
3. **Model Size**: Smaller sizes improve memory usage and storage efficiency.

Changes Observed Across Quantization Approaches

1. Whole Model Quantization

- **Perplexity**: Reduced from 1427.59 to 1044.63 (significant improvement).
- **Latency**: Slightly improved (15.60ms → 15.50ms).
- **Model Size**: Reduced by ~75% (1353.54 MB → 338.39 MB).

2. MLP-Only Quantization

- **Perplexity**: Moderately reduced (1427.59 → 1298.64).
- **Latency**: Marginal improvement (15.71ms → 15.64ms).
- **Model Size**: Reduced by ~42.6% (1353.54 MB → 777.19 MB).

3. Attention-Only Quantization

- **Perplexity**: Increased slightly (1427.59 → 1479.80).
- **Latency**: Increased marginally (15.60ms → 15.91ms).
- **Model Size**: Reduced by ~21.3% (1353.54 MB → 1065.26 MB).

4. Bitsandbytes Library Quantization

Quantization Type	Model Size (MB)	Latency (ms)	Perplexity
Original	1353.54	15.86	1427.59
8-bit	388.77	98.30	1440.05
4-bit	244.77	38.27	1531.03
NF4	244.77	38.16	1358.93

- **8-bit Quantization**: Significant reduction in size (~71%), but latency increased substantially, and perplexity marginally worsened.
- **4-bit Quantization**: Excellent size reduction (~81.9%), but latency worsened, and perplexity increased notably.
- **NF4 Quantization**: Comparable size reduction to 4-bit; latency improved over 4-bit, and perplexity showed noticeable improvement.

Quantitative Insights

- **Whole Model Quantization**: Optimal trade-off between size, latency, and perplexity.
- **MLP-Only Quantization**: Moderate gains in size and perplexity but maintains minimal latency impact.
- **Attention-Only Quantization**: Minimal benefits with increased perplexity and latency.
- **Bitsandbytes Library**:
    - **8-bit**: Best for memory-constrained settings if latency is not critical.
    - **4-bit**: Most efficient for memory but has performance drawbacks.
    - **NF4**: Balanced for size, latency, and perplexity, among the extreme quantization methods.

Qualitative Insights

- Whole model quantization delivers the most balanced results.
- Extreme quantization (e.g., 4-bit) can lead to degraded performance, making it suitable only for specific use cases.

Trade-off Analysis

Component-wise Approaches

1. **Whole Model Quantization**
    - Best overall trade-off
    - Significant memory reduction (-75.0%)
    - Improved perplexity (-26.8%)
    - Minimal latency impact (-0.6%)
2. **MLP-only Quantization**
    - Moderate memory savings (-42.6%)
    - Acceptable perplexity impact (-9.0%)
    - Negligible latency impact (+0.3%)
3. **Attention-only Quantization**
    - Least effective approach
    - Minor memory savings (-21.3%)
    - Slight perplexity degradation (+3.7%)
    - Small latency increase (+2.0%)

Bits and Bytes Approaches

1. **NF4 Quantization**
    - Best bits and bytes trade-off
    - Maximum memory reduction (-81.9%)
    - Improved perplexity (-4.8%)
    - Significant but manageable latency impact (+140.6%)
2. **8-bit Quantization**
    - Good memory savings (-71.3%)
    - Minimal perplexity impact (+0.9%)
    - Highest latency impact (+519.8%)
3. **4-bit Quantization**

- Maximum memory reduction (-81.9%)
- Worst perplexity impact (+7.2%)
- High latency impact (+141.3%)

Recommendations

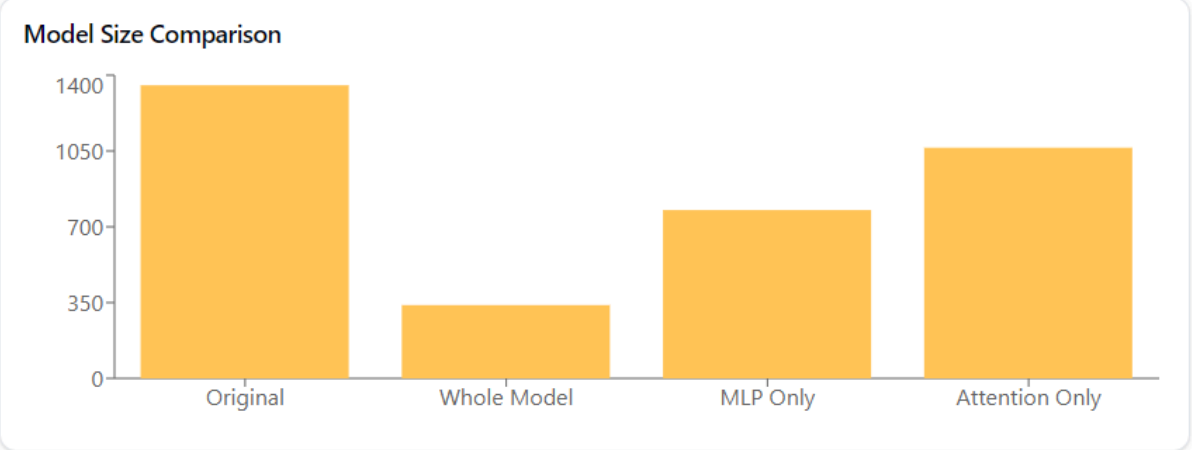
1. **For Maximum Efficiency**
   - Use whole model quantization when possible
   - Provides best balance of memory savings and performance
   - Minimal speed impact
2. **For Maximum Compression**
   - Use NF4 quantization
   - Achieves highest compression with acceptable trade-offs
   - Better perplexity than standard 4-bit
3. **For Quality-Critical Applications**
   - Use whole model quantization or NF4
   - Both show perplexity improvements
   - Choose based on latency requirements
4. **For Speed-Critical Applications**
   - Use component-wise quantization
   - Minimal impact on inference speed
   - Still provides meaningful memory savings

Conclusions

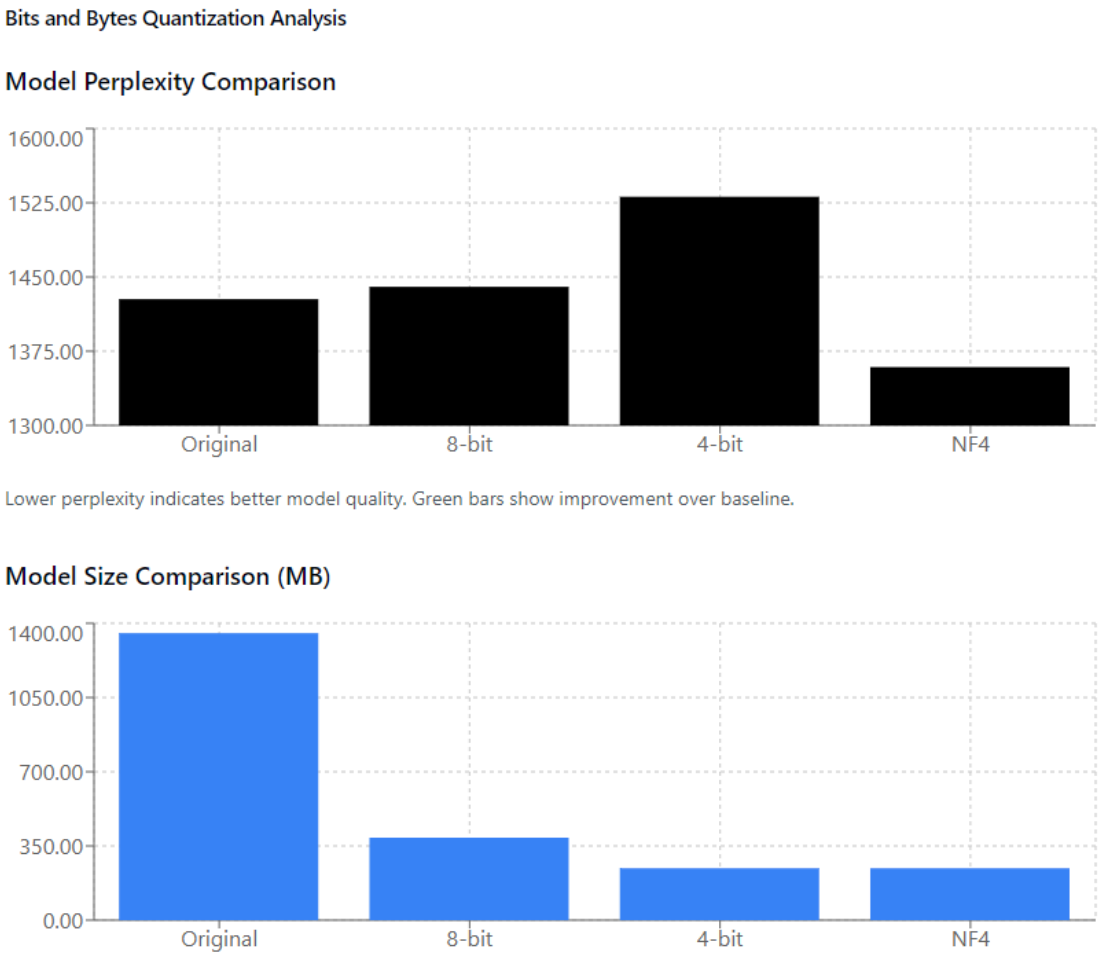
1. Different quantization approaches offer distinct trade-offs:
   - Component-wise methods maintain speed but offer less compression
   - Bits and bytes methods provide maximum compression but impact speed
2. Unexpected findings:
   - Whole model quantization improves perplexity
   - NF4 maintains better quality than higher precision 8-bit in some metrics
3. Best practices:
   - Consider application requirements when choosing quantization method

- Monitor all three metrics (perplexity, latency, size) when evaluating approaches
- Test multiple approaches before deployment





Bitsandbytes results



Understanding NF4 vs Linear Quantization

1. Basic Concepts

Linear Quantization

Linear quantization divides the numerical range into equally-spaced intervals.
For an n-bit quantization:
- Range is divided into 2^n equal intervals
- Each value is rounded to nearest quantization point
- Spacing between levels remains constant
- Example (4-bit): 16 equally spaced levels from min to max

NF4 Quantization

NF4 uses non-uniform intervals based on the normal distribution:

- Smaller intervals near zero (where most weights cluster)
- Larger intervals in the tails
- Optimized for typical neural network weight distributions
- Still uses 4 bits (16 levels) but with variable spacing

	Linear 4-bit	NF4
Spacing	Uniform	Non-uniform
Level placement	Fixed	Data-dependent
Optimization	Range-based	Distribution-based
Precision near 0	Lower	Higher
Tail handling	Equal to center	Coarser

Impact Analysis

Memory Efficiency
Both methods use 4 bits per value, but achieve different compression characteristics:

Linear 4-bit:
- Consistent compression ratio
- Simple implementation
- Higher quantization error for small values

NF4:
- Same compression ratio (4:1)
- More complex implementation
- Better preservation of small values

Model Accuracy Impact

From our empirical results:

Method	Perplexity	Change
Original	1427.59	-
Linear 4-bit	1531.03	+7.2%

NF4	1358.93	-4.8%

Key observations:
1. NF4 actually improves perplexity vs original
2. Linear 4-bit degrades perplexity by 7.2%
3. NF4 provides 11.9% better perplexity than linear 4-bit

Computational Efficiency

Both methods show similar computational characteristics:

Method	Latency (ms)	Change
Original	15.86	-
Linear 4-bit	38.27	+141.3%
NF4	38.16	+140.6%

Observations:
1. Similar latency impact
2. Both require dequantization during inference
3. NF4's complex spacing doesn't significantly affect speed

Practical Implications

Advantages of NF4
1. Better preservation of small weights
   - Critical for model accuracy
   - Reduces information loss
   - Maintains network dynamics

2. Distribution-aware quantization
   - Matches neural network statistics
   - Optimizes bit allocation
   - Reduces quantization error

3. Improved accuracy-compression trade-off
   - Better perplexity than linear
   - Same memory savings
   - Similar computational cost

Limitations

1. More complex implementation
2. Requires distribution analysis
3. Less intuitive than linear quantization

Best Practices

When to Use NF4

1. Large language models
2. Models with critical small weights
3. When accuracy is priority

When to Use Linear Quantization

1. Simple implementations
2. Uniform weight distributions
3. Hardware-constrained scenarios